# Dr. Eng. Raffaele Mineo



## Research Fellow in Artificial Intelligence

Department of Industrial, Electrical, Electronic and Computer Engineering (DIEEI), University of Catania
Catania, Italy

raffaele.mineo[at]phd.unict.it | Google Scholar | ORCID | Scopus | Web of Science
Semantic Scholar | DBLP | OpenReview | GitHub | PeRCeiVe.AI | ResearchGate | LinkedIn | Academia.edu

**Website:** https://IngRaffaeleMineo.github.io/

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*"My Life is The Research, and I am truly lucky to be able to live it to the fullest."*

## Profile

My research covers cardiovascular imaging and virtual physiology, multimodal and radar-based sign language recognition, cognitive assessment, continual and federated learning, vision-language adaptation, and efficient deep learning. I have worked on medical imaging, multimodal sensing and embedded AI across academic, clinical and industrial collaborations.

## Academic positions and research experience

**2024–present** — **Research Fellow**
*University of Catania - DIEEI*

HORIZON Europe NEUROKIT2E: knowledge distillation and meta-learning for on-chip AI inference; medical imaging, multimodal learning, efficient/embedded AI and project research activities.

**2023–present** — **AI consulting and industrial R&D collaboration**
*DeepSensing S.r.l., Catania*

Deep-learning systems for industrial inspection; multi-GPU experimentation; edge and neuromorphic computing; predictive reliability; Horizon-project activities and KPI-oriented validation.

**from 2023** — **Research collaboration**
*STMicroelectronics R&D Power and Discrete, Catania*

Artificial intelligence for modeling and predictive reliability in automotive and power-electronics contexts.

**2019–present** — **Research member / assistant**
*PeRCeiVe.AI Lab, University of Catania*

Computer vision, pattern recognition, medical imaging, multimodal AI and applied deep learning.

## Education

**completed 2026** — **National PhD in Artificial Intelligence - Health and Life Sciences**

*University Campus Bio-Medico of Rome, with University of Catania affiliation*

37th cycle.

**2019–2021** — **M.Sc. in Computer Engineering**
*University of Catania*

110/110 cum laude. Thesis: *Prediction of paroxysmal events through an anomaly detection approach based on Transformers*.

2016–2019 **B.Sc. in Computer Engineering**
*University of Catania*

99/110. Thesis on the design and implementation of an optical data-communication interface for Li-Fi applications.

2016 **Scientific High School Diploma**
*Liceo Scientifico “L. Da Vinci”, Floridia*

97/100.

## Research themes

- **Cardiovascular AI and virtual physiology:** coronary angiography, FFR/iFR estimation, spatio-temporal modeling, clinically meaningful representations and translational validation.
- **Multimodal and privacy-aware sensing:** radar, RGB, depth, skeleton and multimodal learning for sign language recognition and accessible healthcare communication.
- **Adaptive and distributed learning:** personalized continual/incremental learning, self-supervision, federated learning and prompt-based adaptation.
- **Efficient and embedded AI:** knowledge distillation, post-training quantization, edge inference, STM32/embedded deployment and resource-constrained optimization.
- **Applied machine intelligence:** cognitive assessment, industrial inspection, predictive reliability, anomaly detection, volcanology and data-driven engineering systems.

## Research trajectory

2026 **Model compression & efficient AI**

Knowledge distillation, post-training and mixed-precision quantization, sparsification and pruning for efficient vision models and deployment under compute constraints.

2025 **Prompt-based adaptation & federated/continual learning**

Vision-language prompt tuning, personalized continual/incremental learning and federated learning for data-efficient medical image analysis.

2024 **Digital cognitive assessment**

Computer vision for cognitive screening, including facial micro-movement analysis and eye-gaze modeling for automated neurocognitive assessment.

2023–2026 **Italian Sign Language (LIS) & privacy-aware multimodal sensing**

Multimodal LIS recognition for healthcare communication using RGB, depth, skeleton and radar sensing; MultiMedaLIS, radar-based recognition and SignEval benchmarks/challenges.

2022–2023 **Cardiovascular imaging & computational physiology**

Coronary angiography, stenosis assessment and non-invasive FFR/iFR estimation, establishing the methodological foundations of the STARFLOW research line.

2020–2021 **Volcanic seismology & anomaly detection**

Deep learning for seismic anomaly detection and volcanic-unrest analysis with INGV/EUROVOLC, including the VolUnD toolkit and machine-learning database activities.

2020 **Thoracic imaging & lung analysis**

Deep-learning and computer-vision methods for lung and thoracic imaging, extending early medical-imaging work beyond neuroimaging.

2019–2020 **Brain signal analysis & neuroimaging (EEG/fMRI)**

Early research on brain data and representation learning across EEG and fMRI, with emphasis on spatio-temporal patterns and neuroimaging analysis.

## Patents, inventions and technology transfer

2026

**Angiographic image processing method and system**
*Italian patent no. 102024000014434; PCT/IB2025/056252; WO 2025/262634 A1*

Italian patent granted by UIBM on 20 July 2026. The patent family protects AI-based angiographic processing underpinning the STARFLOW line of research.

# Academic service and leadership

### Editorial roles

- Associate Editor - Frontiers in Communication (2026–)
- Lead Guest Editor - Cognitive Computation, Special Issue “Cognitive and Multimodal Sign Language Processing: Recognition, Understanding, Translation, Generation, and Inclusive Interaction” (2026)

### Area-chair, program and organizing roles

- Area Chair - MICCAI 2026
- Organizing Committee - IEEE ISBI 2026, Associate to Technical Program Chairs
- Session Chair - European Conference on EDGE AI Technologies and Applications (EEAI) 2026
- Area Chair - IJCNN 2027
- Area Chair - IEEE International Conference on Automatic Face and Gesture Recognition (FG) 2027
- Area Chair - IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2027
- Area Chair - IEEE International Symposium on Biomedical Imaging (ISBI) 2027
- Organizing Committee - IEEE ISBI 2027, Associate to Technical Program Chairs
- Paper Selection Committee - IPMI 2027

### Workshops, challenges and special sessions

- Organizer - ICCV 2025: 1st Workshop on Multimodal Sign Language Recognition (MSLR 2025), featuring the SignEval 2025 Challenge
- Organizer - ACM Multimedia 2025: (PILM ’25) International Workshop on Personalized Incremental Learning in Medicine
- Organizer - IEEE ISBI 2026 Special Session: “Privacy-Aware, Data-Efficient AI via Personalized Incremental and Federated Learning in Healthcare”
- Workshop Chair - CVPR 2026: 2nd Workshop on Multimodal Sign Language Recognition (MSLR), featuring SignEval 2026: The Second Multimodal Sign Language Recognition Challenge

### Journal reviewing

- Computer Vision and Image Understanding (CVIU), since 2023
- Machine Vision and Applications (MVAP), since 2023
- ACM Transactions on Intelligent Systems and Technology (TIST), since 2024
- IEEE Transactions on Medical Imaging (TMI), since 2025
- Nature Communications, since 2025
- IEEE Open Journal of the Computer Society (OJCS), since 2025
- Pattern Recognition, since 2026
- IET Cyber-Systems and Robotics, since 2026
- BioData Mining, since 2026
- BMC Medical Informatics and Decision Making, since 2026
- Disability and Rehabilitation: Assistive Technology, since 2026
- Frontiers in Digital Health, since 2026
- Frontiers in Aging Neuroscience, since 2026
- Frontiers in Medicine, since 2026
- International Journal of Cognitive Computing in Engineering (IJCCE), since 2026
- Journal of Imaging Informatics in Medicine (JIIM), since 2026
- IEEE Transactions on Biomedical Engineering (TBME), since 2026

### Conference and workshop reviewing

- **Conferences:** ECCV 2024; BIBM 2024; RTSI 2024; ISBI 2026; MICCAI 2026; IJCNN 2026; EMBC 2026; NeurIPS 2026; AAAI 2027.
- **Workshops:** MICCAI AIPAD 2024; IJCNN SPAN 2025; CVPR MSLR 2026; ICPR CVBMC 2026; MICCAI AFRICAI

2026.

### Volunteering, mentoring and research advising

- Mentor - GreenMindAI Catania Hackathon 2025
- Student Volunteer - 33rd ACM International Conference on Multimedia (ACM Multimedia 2025)
- Proctor - IEEEXtreme 20.0 (2026)
- Proctor Ambassador - IEEEXtreme 20.0 (2026)
- Research Advisor - RISE-MICCAI 2027 Publication Accelerator
- Mentor - Open Hackathons

## Awards, grants and recognition

**2025** **ICCV 2025 Broadening Participation (BP) Support**
*IEEE/CVF International Conference on Computer Vision (ICCV 2025)*

Recipient of support through the ICCV 2025 Broadening Participation program, which provides travel and/or registration support.

**2025** **ACM Multimedia 2025 Student Travel Award**
*33rd ACM International Conference on Multimedia (ACM Multimedia 2025)*

Student travel award from ACM SIGMM / the ACM Multimedia 2025 organizing committee.

## Publications and research outputs by topic

Publications and research outputs are grouped by their primary research topic.

### Cardiovascular AI and Computational Physiology

**2025** **Non-invasive physiological assessment of intermediate coronary stenoses from plain angiography through artificial intelligence: the STARFLOW system** *
Ovidio De Filippo*, **Raffaele Mineo***, Michele Millesimo, Wojciech Wańha, Federica Proietto Salanitri, Antonio Greco, Antonio Maria Leone, Luca Franchin, Simone Palazzo, Giorgio Quadri, Domenico Tuttolomondo, Enrico Fabris, Gianluca Campo, Alessandra Truffa Giachet, Francesco Bruno, Mario Iannaccone, Giacomo Boccuzzi, Nicola Gaibazzi, Ferdinando Varbella, Wojciech Wojakowski, Michele Maremmani, Guglielmo Gallone, Gianfranco Sinagra, Davide Capodanno, Giuseppe Musumeci, Paolo Boretto, Pawel Pawlus, Andrea Saglietto, Francesco Burzotta, Marco Aldinucci, Daniela Giordano, Gaetano Maria De Ferrari, Concetto Spampinato, Fabrizio D'Ascenzo
*European Heart Journal-Quality of Care and Clinical Outcomes*

**2024** **A Convolutional-Transformer Model for FFR and iFR Assessment From Coronary Angiography**
**Raffaele Mineo**, F Proietto Salanitri, G Bellitto, I Kavasidis, O De Filippo, M Millesimo, GM De Ferrari, M Aldinucci, D Giordano, S Palazzo, F D'Ascenzo, C Spampinato
*IEEE Transactions on Medical Imaging*

**2024** **Coronary Artery Stenosis Assessment in X-Ray Angiography Through Spatio-Temporal Attention for Non-Invasive FFR and iFR Estimation**
**Raffaele Mineo**, Federica Proietto Salanitri, Giovanni Bellitto, Ovidio De Filippo, Fabrizio D'Ascenzo, Simone Palazzo, Concetto Spampinato
*Proceedings of the 17th International Joint Conference on Biomedical Engineering Systems and Technologies - Volume 1: BIOIMAGING*

**2023** **FeDETR: A Federated Approach for Stenosis Detection in Coronary Angiography**
**Raffaele Mineo**, Amelia Sorrenti, Federica Proietto Salanitri
*International Conference on Image Analysis and Processing*

### Multimodal Sensing and Inclusive Healthcare

**2026** **SignEval 2026 Challenge Results** *
Ahmed Abul Hasanaath, **Raffaele Mineo**, Hamzah Luqman, Sarah Alyami, Maad Alowaifeer, Amelia Sorrenti, Gaia Caligiore, Sabina Fontana, Egidio Ragonese, Giovanni Bellitto, Federica Proietto Salanitri, Concetto Spampinato, Motaz Alfarraj, Mufti Mahmud, Simone Palazzo, Nour Imane Zeghib
*IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*

2026 **A Benchmark for Radar-Based Italian Sign Language Recognition using Frequency-Domain Range-Time Maps**
**Raffaele Mineo**, Amelia Sorrenti, Gaia Caligiore, Federica Proietto Salanitri, Giovanni Bellitto, Senya Polikovsky, Sabina Fontana, Egidio Ragonese, Concetto Spampinato, Simone Palazzo
*IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*

2025 **The SignEval 2025 Challenge at the ICCV Multimodal Sign Language Recognition Workshop: Results and Discussion** *
Hamzah Luqman*, **Raffaele Mineo***, Murtadha Aljubran, Ahmed Abul Hasanaath, Amelia Sorrenti, Sarah Alyami, Sadam Al-Azani, Maad Alowaifeer, JiHwan Moon, Vaclav Javorek, Tomas Zelezny, Marek Hruz, Gaia Caligiore, Silvio Giancola, Senya Polikovsky, Motaz Alfarraj, Sabina Fontana, Mufti Mahmud, Muhammad Haris Khan, Kamrul Islam, Sevgi Gurbuz, Egidio Ragonese, Giovanni Bellitto, Federica Proietto Salanitri, Concetto Spampinato, Simone Palazzo
*Proceedings of the IEEE/CVF International Conference on Computer Vision*

2025 **Text-Aligned Radar-Based Sign Language Recognition for Healthcare Communication**
**Raffaele Mineo**, Amelia Sorrenti, Gaia Caligiore, Federica Proietto Salanitri, Giovanni Bellitto, Senya Polikovsky, Sabina Fontana, Egidio Ragonese, Concetto Spampinato, Simone Palazzo
*Proceedings of the IEEE/CVF International Conference on Computer Vision*

2025 **Radar-Based Imaging for Sign Language Recognition in Medical Communication**
**Raffaele Mineo**, Gaia Caligiore, Federica Proietto Salanitri, Isaak Kavasidis, Senya Polikovsky, Sabina Fontana, Egidio Ragonese, Concetto Spampinato, Simone Palazzo
*International Conference on Medical Image Computing and Computer-Assisted Intervention*

2024 **Sign Language Recognition for Patient-Doctor Communication: A Multimedia/Multimodal Dataset**
**Raffaele Mineo**, Gaia Caligiore, Concetto Spampinato, Sabina Fontana, Simone Palazzo, Egidio Ragonese
*2024 IEEE 8th Forum on Research and Technologies for Society and Industry Innovation (RTSI)*

2024 **Multisource Approaches to Italian Sign Language (LIS) Recognition: Insights from the Multi-MedaLIS Dataset** *
Gaia Caligiore*, **Raffaele Mineo***, Concetto Spampinato, Egidio Ragonese, Simone Palazzo, Sabina Fontana
*2024 10th Italian Conference on Computational Linguistics (CLiC-it)*

## Neurocognitive and Brain Signal AI

2025 **Automated MoCA Score Estimation Using Eye-Gaze Data and Vision Transformers**
**Raffaele Mineo**, Isaak Kavasidis, Federica Proietto Salanitri, Lisa Passarello, Giovanni Piccininno, Vincenzo Masciale, Alessandro Anselmo, Cristiano Convertino, Domenico Rotondi, Nicola Laurieri, Simone Palazzo, Concetto Spampinato, Manuela Pennisi, Daniela Giordano
*2025 IEEE 38th International Symposium on Computer-Based Medical Systems (CBMS)*

2024 **Magnifying facial micro-movements for cognitive evaluation**
**Raffaele Mineo**, Federica Proietto Salanitri, Lisa Passarello, Alberto Sardella, Laura Messina, Manuela Pennisi, Daniela Giordano, Simone Palazzo, Concetto Spampinato
*2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*

2023 **Ensemble and personalized transformer models for subject identification and relapse detection in e-prevention challenge**
Salvatore Calcagno, **Raffaele Mineo**, Daniela Giordano, Concetto Spampinato
*ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*

2023 **Dynamic Graph Attention: Unraveling Spatio-Temporal Synchrony in EEG Data**
Federica Proietto Salanitri, Giovanni Bellitto, **Raffaele Mineo**, Matteo Pennisi, Amelia Sorrenti, Salvatore Calcagno, Daniela Giordano, Simone Palazzo, Concetto Spampinato
*2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*

## Adaptive, Efficient and Edge AI

2025 **Timely Predicting Power Mosfet Failure in EV: AI Based Metodology and Deployment at the Edge**
Robin Faro, Francesco Cancelliere, Alesssandro Strano, **Raffaele Mineo**
*2025 IEEE Symposia on Computational Intelligence for Energy, Transport and Environmental Sustainability (CIETES Companion)*

2025 **Learning Joint Text and Visual Tokens in CLIP for Medical Image Analysis**
**Raffaele Mineo**, Giovanni Bellitto, Federica Proietto Salanitri, Rutger Hendrix, Concetto Spampinato, Simone Palazzo
*Proceedings of the 2nd International Workshop on Multimedia Computing for Health and Medicine*

2025 **HC-PTQ: Poincaré-Based Hyperbolic Clustering for Data-Free Quantization of Vision Transformers**
**Raffaele Mineo**, Simone Palazzo, Concetto Spampinato, Francesco Rundo
*Proceedings of the IEEE/CVF International Conference on Computer Vision*

2025 **Conditional Visuo-Textual Prompt Learning for Medical Image Analysis**
**Raffaele Mineo**, Giovanni Bellitto, Federica Proietto Salanitri, Rutger Hendrix, Ulas Bagci, Daniela Giordano, Concetto Spampinato, Simone Palazzo
*2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI)*

### Scientific and Industrial Intelligent Systems

**2026** **When a model is not Enough: A Complementary AI Pipeline for Ultra-Safe PCBA Defect Detection**
Alberto Faro, Francesco Cancelliere, Robin Faro, **Raffaele Mineo**
*The Autonomous Edge - Intelligence Embedded in Industrial Applications, River Publishers, pp. 129–149, DOI: 10.1201/9788743815273-8 (EEAI 2025 book chapter)*

**2025** **PCB-SAID: A Low-Cost Camera-Based Dataset for Few-Shot SMD Assembly Inspection**
**Raffaele Mineo**, Amelia Sorrenti, Robin Faro, Gabriele Mineo, Francesco Cancelliere, Alberto Faro
*Proceedings of the IEEE/CVF International Conference on Computer Vision*

**2023** **BioTrak: A Blockchain-based Platform for Food Chain Logistics Traceability**
Alessia Spitalleri, Isaak Kavasidis, Vincenzo Cartelli, **Raffaele Mineo**, Francesco Rundo, Simone Palazzo, Concetto Spampinato, Daniela Giordano
*2023 International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS)*

**2021** **VOLcano UNrest Detection (VolUnD) Toolkit**
**R. Mineo**, P. Spadaro, F. Cannavò, S. Palazzo, A. Cannata, C. Spampinato
*Zenodo*

**2021** **EUROVOLC - Deliverable Report - D9.3 Machine learning tool kit and database**
S. Palazzo, **R. Mineo**, C. Spampinato, F. Cannavò, A. Cannata, C. Montagna, J. M. Saurel, O. Geber, A. Lemarchand
*HORIZON 2020 EUROVOLC Project*

### Research Synthesis and Scholarly Outputs

**2027** **Advancing Healthcare with Multimodal AI: Enhanced Diagnosis, Cognitive Monitoring, and Inclusive Medical Communication**
**Raffaele Mineo**
*Medical Image Computing and Computer Assisted Intervention – MICCAI 2025 Doctoral Symposium; Lecture Notes in Computer Science, vol. 16489 (Springer)*

**2025** **Proceedings of the International Workshop on Personalized Incremental Learning in Medicine (PILM '25)**
Simone Palazzo, Giovanni Bellitto, **Raffaele Mineo**, Amelia Sorrenti, Camila Gonzalez, Angelo Porrello, Lorenzo Bonicelli
*ACM Multimedia 2025*

* Joint first author where explicitly indicated in the supplied publication list.

## Research projects and collaborations

- **HORIZON Europe - NEUROKIT2E** (Grant Agreement 101112268): research on knowledge distillation and meta-learning for on-chip AI inference; contributions also spanning embedded/edge AI and performance evaluation.
- **HORIZON Europe - EdgeAI related activities:** deep-learning algorithms for defect detection on electronic boards and wood products; multi-GPU and edge-computing experimentation; neuromorphic approaches; KPI and functional/non-functional requirement evaluation.
- **HORIZON 2020 - EUROVOLC** (Grant Agreement 731070): machine-learning toolkit and database for volcanic-unrest analysis; VolUnD toolkit and D9.3 deliverable.
- Research collaborations with the **Istituto Nazionale di Geofisica e Vulcanologia (INGV), Catania**; **AOU Città della Salute e della Scienza, Turin**; **Max Planck Institute for Intelligent Systems, Tübingen**; and **STMicroelectronics R&D Power and Discrete, Catania**.
- Industrial R&D / consulting collaboration with **DeepSensing S.r.l., Catania**.

## Teaching

- Teaching Tutor, **Database and Web Programming**, Computer Engineering, University of Catania - multiple editions, 2022–2025.
- Teaching Tutor, **Fundamentals of Computer Science**, Industrial Engineering, University of Catania - 2022–2023.

## Professional memberships and technical communities

- CVPL - Associazione Italiana per la ricerca in Computer Vision, Pattern Recognition e Machine Learning (since 2019)
- IEEE Engineering in Medicine and Biology Society (EMBS), since 2024

- IEEE Signal Processing Society (SPS), since 2025
- Association for Computing Machinery (ACM), since 2025
- ACM Special Interest Group on Multimedia (SIGMM), since 2025
- IEEE Computer Society Technical Community on Pattern Analysis and Machine Intelligence (TCPAMI), since 2026
- IEEE Engineering in Medicine and Biology Society (EMBS) Technical Committee on Biomedical Imaging and Image Processing (BIIP), since 2026

## Technical competencies

- **Artificial intelligence and data:** AI for Medicine, machine learning, deep learning, data acquisition systems, PyTorch, Keras, TensorFlow, R, SQL, MySQL, PostgreSQL.
- **Programming:** Python, C, C++, C#, Java, JavaScript, TypeScript, Go, PHP, AutoIt, OpenGL, QML, Qt Framework / Qt Quick, ASP.NET, Angular, Laravel, Node.js.
- **Embedded systems and sensing:** STM32 firmware, Arduino, Infineon Radar SDK, Intel RealSense SDK, Microsoft Kinect SDK, Stereolabs ZED SDK, edge computing, multi-GPU systems.
- **Engineering and development tools:** Android Studio, Atollic TrueSTUDIO, Eclipse, IntelliJ IDEA, Qt Creator, Visual Studio, Visual Studio Code, AutoCAD, Autodesk EAGLE, Autodesk Tinkercad.
- **Web, systems and networks:** HTML, CSS, SCSS, JSON, client/server applications, network architecture, operating systems.
- **Media, graphics and productivity:** Adobe Illustrator, Photoshop, Acrobat, After Effects, Premiere Pro, Blender, CorelDRAW, Unity, Microsoft Office, Excel and PowerPoint.

## Languages

- Italian - native
- English - B1 (Cambridge certified)
- French - A1
- Spanish - A1

## Professional qualifications, certifications and training

- Cambridge English A2 Key (2012); Cambridge English B1 Preliminary (2014)
- ECDL Core AICA (2014)
- Italian State Examination and professional qualification as Information Engineer (2021)
- Würth Elektronik Italia - Power Expert technical training (2021)
- Mensa Italia certificate / membership credential
- Coursera / DeepLearning.AI: Neural Networks and Deep Learning; Improving Deep Neural Networks; Structuring Machine Learning Projects; Convolutional Neural Networks; Sequence Models; Deep Learning Specialization
- Coursera: AI for Medical Diagnosis; AI for Medical Prognosis; AI for Medical Treatment; AI for Medicine Specialization; Detecting COVID-19 with Chest X-Ray using PyTorch
- Coursera / DeepLearning.AI: Build Basic GANs; Build Better GANs; Apply GANs; Generative Adversarial Networks Specialization

## Additional hands-on technical background

Electronic-board repair for household appliances; desktop/laptop/smartphone/tablet hardware and software repair; installation and maintenance experience with television systems, alarm systems, video surveillance, structured cabling and selected plumbing/HVAC systems; electro-welding and carpentry; preparation of technical-estimative reports and IT appraisals.